PHYS 365/369 — Midterm Fall 2022 — Worked Solutions

16 questions. Every step uses the formula sheet printed at the back of the F2022 midterm. That sheet is older and shorter than the current one. Anything that is *not* on it is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key and the instructor's handwritten written-response solutions.

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	5	D	9	C	13	A
2	C	6	B	10	C	14	D
3	B	7	C	11	A*	15	B
4	C	8	D	12	C	16	B

NOTE ON THE ANSWER KEY

*Q11: the key's 608 nm only comes out if the film thickness is $L = 285$ nm. The table prints 385 nm, which gives 493 nm (not an option). This is very likely a typo in the table. See the full solution.

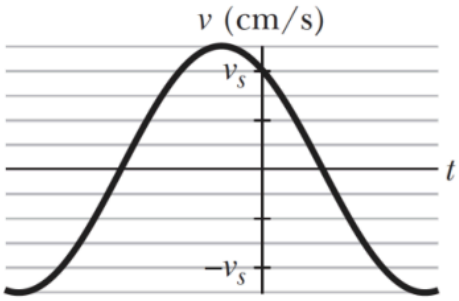
REMEMBER — not on the formula sheet

- SHM: $x = x_m \cos(\omega t + \phi),\ v = -\omega x_m \sin(\omega t + \phi),\ a = -\omega^2 x$. This sheet only gives ω formulas (Q1, Q2).
- Choosing between the two solutions of $\sin^{-1}$: use the sign of the slope or acceleration to pick the right one (Q1).
- $\sin(\theta + \pi) = -\sin\theta$ (Q3).
- In a longitudinal wave, particles oscillate along the direction of travel between $-A$ and $+A$ (Q4).
- The tension in a hanging rope at height x above the bottom equals the weight below that point: $\tau = \mu g x$ (plus any hanging mass). Tension from a hanging block: $\tau = mg$ (Q6, Q14).
- Small angles: $y = D\tan\theta \approx D\sin\theta$ (Q8).
- Number of maxima around two sources: $|m| \leq \frac{d}{\lambda}$. Each $|m| < \frac{d}{\lambda}$ gives 2 points, and $|m| = \frac{d}{\lambda}$ gives 1 point on each side (Q10).
- Reflection phase shift: off a **higher** n gives $\frac{1}{2}\lambda$. Off a **lower** n gives 0 (Q11, Q12).
- Thin film, transmitted rays (extra path $2L$), with $\lambda_n = \dfrac{\lambda}{n_2}$. No net shift: maxima $2L = m\lambda_n$, minima $2L = \left(m - \frac{1}{2}\right)\lambda_n$. Net $\frac{1}{2}\lambda$ shift: the conditions swap (Q11, Q12).
- Zero reflection means $Z_1 = Z_2$ (Q13).
- Speed of a point moving in a circle: $v = r\omega$ (Q16).

Question 1 Phase constant from a velocity graph

1. What is the phase constant for the harmonic oscillator with the velocity function $v(t)$ given in the Figure, if the position function $x(t)$ has the form $x = x_m \cos(\omega t + \phi)$? The vertical axis scale is set by $v_s = 8.0\ cm/s$.

A) $+5.36$ rad
B) $+0.927$ rad
C) $+4.07$ rad
D) -4.07 rad



FROM THE FORMULA SHEET
$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$, $\sin^{-1}(x) = \{\theta,\ \pi - \theta\} + 2\pi n$ (math section)

REMEMBER — not on the formula sheet
From $x = x_m \cos(\omega t + \phi)$: $v = -\omega x_m \sin(\omega t + \phi) = -v_m \sin(\omega t + \phi)$ and $a = -\omega^2 x_m \cos(\omega t + \phi)$.

Step 1. Read the graph. The gridlines are $2\ \frac{\text{cm}}{\text{s}}$ apart ($v_s = 8$ is 4 lines up). The peak is 5 lines up, so $v_m = 10\ \frac{\text{cm}}{\text{s}}$. At $t = 0$, $v = +8\ \frac{\text{cm}}{\text{s}}$ and the curve is going **down**.

Step 2. At $t = 0$:

$$8 = -10\sin\phi \ \Rightarrow\ \sin\phi = -0.8 \ \Rightarrow\ \phi = -0.927\ \text{rad} \ \text{ or } \ \phi = \pi + 0.927 = 4.07\ \text{rad}$$

Step 3. Pick the right one. v is decreasing at $t = 0$, so $a(0) < 0$. Since $a(0) = -\omega^2 x_m \cos\phi$, we need $\cos\phi > 0$.

$\cos(-0.927) = +0.6$ ✓ and $\cos(4.07) = -0.6$ ✗. So $\phi = -0.927$ rad, which is the same angle as:

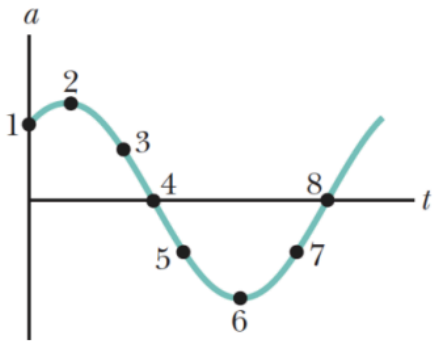
$$\phi = -0.927 + 2\pi = +5.36\ \text{rad}$$

Final answer: **(A)** $+5.36$ **rad** ✓ **Matches answer key**

Question 2 Velocity from the acceleration graph

2. The acceleration $a(t)$ of a particle undergoing SHM is graphed in the Figure. Which of the. following statements is **true** about point **4**?

A) The velocity of the particle is negative
B) The particle is at $-x_m$ with velocity zero
C) The velocity of the particle is positive
D) The particle is at $+x_m$ with velocity zero



REMEMBER — not on the formula sheet
$a = -\omega^2 x$. The velocity is the slope of $x(t)$.

Step 1. Since $x = -\dfrac{a}{\omega^2}$, the graph of $x(t)$ is the $a(t)$ graph flipped upside down.

Step 2. At point 4, a crosses zero going **downward** (positive to negative). So x crosses zero going **upward** (negative to positive).

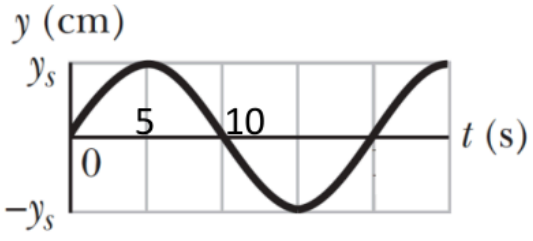
Step 3. x is increasing, so the slope is positive and $v > 0$. It is at $x = 0$, not at $\pm x_m$, so B and D are wrong.

Final answer: **(C) The velocity is positive** ✓ Matches answer key

Question 3 Writing a wave function from a $y(t)$ graph

3. A sinusoidal transverse wave of wavelength 20 cm travels along a string in the positive direction of an x axis. The displacement y of the string particle at $x = 0$ is given in Figure as a function of time t. The scale of the vertical axis is set by $y_s = 10$ cm. The wave function is to be in the form of:

A) $y(x,t) = +(10\ cm)\sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$
B) $y(x,t) = -(10\ cm)\sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$
C) $y(x,t) = +(10\ cm)\sin\left(\frac{\pi}{20}x - \frac{\pi}{10}t\right)$
D) $y(x,t) = -(10\ cm)\sin\left(\frac{\pi}{20}x - \frac{\pi}{10}t\right)$



FROM THE FORMULA SHEET
$y(x,t) = A\sin(kx \pm \omega t + \phi_0), \quad k = \dfrac{2\pi}{\lambda}, \quad \omega = \dfrac{2\pi}{T}$

REMEMBER — not on the formula sheet
$\sin(\theta + \pi) = -\sin\theta$

Step 1. Wave number: $k = \dfrac{2\pi}{20\text{ cm}} = \dfrac{\pi}{10}$ rad per cm. That rules out C and D, which have $\frac{\pi}{20}$.

Step 2. Period from the graph: it crosses zero at $t = 0$ and at $t = 10$ s, which is half a period, so $T = 20$ s and $\omega = \dfrac{2\pi}{20} = \dfrac{\pi}{10}\ \frac{\text{rad}}{\text{s}}$.

Step 3. For a wave moving in $+x$, use $y = 10\sin(kx - \omega t + \phi_0)$. At $x = 0$: $y(0,t) = 10\sin(-\omega t + \phi_0)$.

The graph shows $y(0,t) = +10\sin(\omega t)$, which starts at 0 and goes up. Pick ϕ_0 to match:

$$10\sin(-\omega t + \pi) = 10\sin(\omega t)\ \checkmark \quad \Rightarrow \quad \phi_0 = \pi$$

Step 4. So:

$$y = 10\sin\left(\tfrac{\pi}{10}x - \tfrac{\pi}{10}t + \pi\right) = -(10\text{ cm})\sin\left(\tfrac{\pi}{10}x - \tfrac{\pi}{10}t\right)$$

Final answer: **(B)** $y = -(10\text{ cm})\sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$ ✓ **Matches answer key**

Question 4 Motion of particles in a longitudinal wave

4. A longitudinal wave with an amplitude of 0.02 m moves horizontally along a Slinky with a speed of 2 m/s. Which one of the following statements concerning this wave is true?

   A) Each particle in the Slinky moves a horizontal distance of 2 m each second.
   B) Each particle in the Slinky moves a vertical distance of 0.04 m during each period of the wave.
   C) Each particle in the Slinky moves a horizontal distance of 0.04 m during each period of the wave.
   D) Each particle in the Slinky moves a horizontal distance of 0.02 m during each period of the wave.

Step 1. In a longitudinal wave each particle oscillates **horizontally** (along the direction of travel) between -0.02 m and $+0.02$ m. That excludes B (vertical).

Step 2. The span of that motion is $2A = 0.04$ m. A describes the wave speed, not the particle motion. D gives only one amplitude, not the full back-and-forth span.

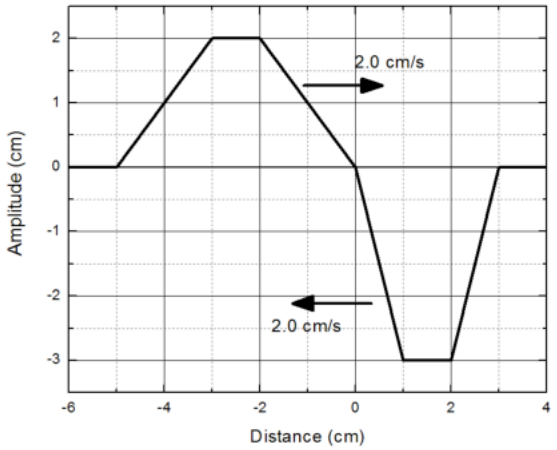
NOTE ON THE ANSWER KEY
Strictly, the total path a particle travels in one period is $4A = 0.08$ m (out and back on both sides). The key's intended meaning for C is that the particle moves across a horizontal range of $2A = 0.04$ m. C is the intended answer.

Final answer: **(C) Horizontal distance of 0.04 m** ✓ **Matches answer key**

Question 5 Superposing two moving pulses

5. The graph shows two waves at time $t = 0$ s, one moving toward the right at 2.0 cm/s and the other moving toward the left at 2.0 cm/s. What will the amplitude be at $x = 0$ at time $t = 0.5$ s?

A) -3 cm
B) zero cm
C) -1 cm
D) -2 cm



FROM THE FORMULA SHEET

$$y(x,t) = f(x \pm vt)$$

Step 1. In $t = 0.5$ s each pulse moves $\Delta x = (2.0)(0.5) = 1.0$ cm.

Step 2. The right-moving pulse brings to $x = 0$ whatever it had at $x = -1$ cm. On the line from $(-2, 2)$ to $(0, 0)$, that value is $+1$ cm.

Step 3. The left-moving pulse brings to $x = 0$ whatever it had at $x = +1$ cm. On its flat bottom, that value is -3 cm.

Step 4. Superposition:

$$y(0,\ 0.5\text{ s}) = (+1) + (-3) = -2\text{ cm}$$

Final answer: **(D) -2 cm** ✓ **Matches answer key**

Question 6 Wave speed up a hanging rope

6. A climbing rope is hanging from the ceiling in a gymnasium. A student grabs the end of the rope and begins moving it back and forth with a constant amplitude and frequency. A transverse wave moves up the rope. Which of the following statements describing the speed of the wave is true?

A) The speed of the wave decreases as it moves upward.
B) The speed of the wave increases as it moves upward.
C) The speed of the wave is constant as it moves upward.
D) The speed of the wave does not depend on the mass of the rope.

FROM THE FORMULA SHEET

$$v = \sqrt{\frac{F_T}{\mu}}$$

REMEMBER — not on the formula sheet
At height x above the bottom end, the rope holds up the weight of everything below it: $F_T = \mu x g$ (plus the student's pull on the end).

Step 1. The tension grows as you go up, because more rope hangs below each point. μ is the same everywhere.

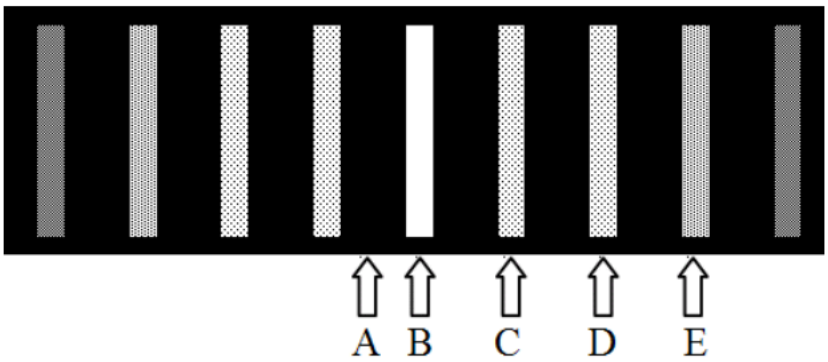
Step 2. So $v = \sqrt{\dfrac{F_T}{\mu}}$ increases going up. For a free rope, $v = \sqrt{\dfrac{\mu x g}{\mu}} = \sqrt{gx}$, which gets bigger as x increases.

Final answer: **(B) Speed increases moving upward** ✓ Matches answer key

Question 7 Which fringe has a 1320 nm path difference?

7. A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled "**B.**" Which one of these fringes is produced when the path difference is 1320 nm?

A) A
B) C
C) D
D) E



FROM THE FORMULA SHEET
$\Delta = |r_2 - r_1| = m\lambda$ (bright fringes)

$$m = \frac{\Delta}{\lambda} = \frac{1320}{660} = 2$$

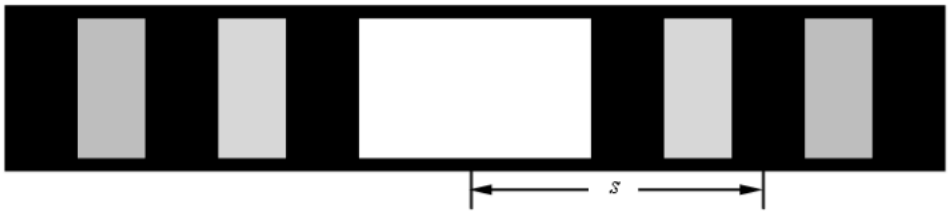
Counting bright fringes from the centre: B is $m = 0$, C is $m = 1$, D is $m = 2$. (A is the dark fringe next to the centre.)

Final answer: **(C) Fringe D** ✓ Matches answer key

Question 8 Single-slit pattern, finding s

8. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2 m from the slits. Determine the distance s.

A) 0.002 m
B) 0.003 m
C) 0.004 m
D) 0.006 m



FROM THE FORMULA SHEET
$\Delta = a \sin\theta = m\lambda$ (single-slit minima)

REMEMBER — not on the formula sheet
Small angles: $y = D\tan\theta \approx D\sin\theta$, so $y_m = \dfrac{m\lambda D}{a}$.

Step 1. In the figure, s runs from the centre of the pattern to the **second** dark fringe, so $m = 2$.

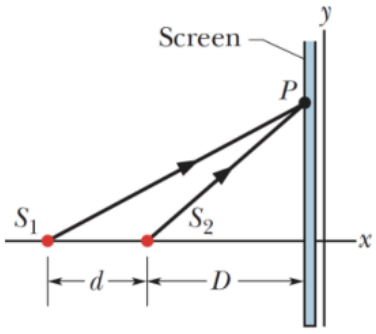
$$s = y_2 = \frac{2\lambda D}{a} = \frac{2(600 \times 10^{-9})(2)}{4 \times 10^{-4}} = \frac{2.4 \times 10^{-6}}{4 \times 10^{-4}} = 6 \times 10^{-3}\ \text{m}$$

Final answer: **(D) 0.006 m** ✓ Matches answer key

Question 9 Maximum path difference on the screen

9. In the Figure, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 6.00\,\lambda$ on an x axis. A viewing screen is at distance $D = 20.0\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P. What multiple of λ gives the maximum phase difference between two rays on the screen?

A) Zero
B) 20.0
C) 6.00
D) 26



FROM THE FORMULA SHEET
$\Delta = |r_2 - r_1|$

Step 1. For a point P on the screen, $\Delta = r_1 - r_2$. By the triangle inequality, $r_1 \le r_2 + d$, so $\Delta \le d$.

Step 2. Equality holds when P is on the x-axis ($y_P = 0$), where both rays travel along the same line:

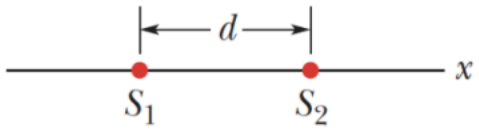
$$\Delta_{\max} = d = 6.00\,\lambda$$

Final answer: **(C) 6.00** ✓ Matches answer key

Question 10 Counting maxima around two sources

10. In the Figure, two radiofrequency point sources S_1 and S_2, separated by distance $d = 1.0$ m, are radiating in phase with $\lambda = 0.50$ m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

A) 3
B) 5
C) 8
D) 10



FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$

Step 1. $\sin\theta = \dfrac{m\lambda}{d} = \dfrac{m(0.50)}{1.0} = \dfrac{m}{2}$. We need $|\sin\theta| \le 1$, so $m = 0, \pm1, \pm2$.

Step 2. Going around a full circle: $m = 0$ gives 2 points (above and below), $m = \pm1$ gives 4 points, and $m = \pm2$ gives 2 points (on the axis, one on each side).

$$2 + 4 + 2 = 8$$

Final answer: **(C) 8** ✓ Matches answer key

Question 11 Thin film, minimum in transmission

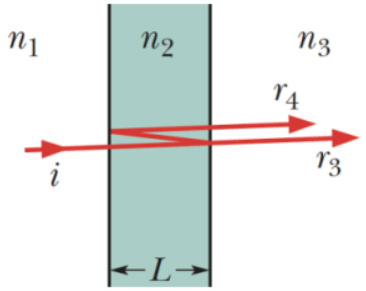
Questions 11 and 12 are related to the following situation:

In the Figure, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3(the light does not reflect inside material 2) and r_4(the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.

Question 11 find the wavelength that is in the visible range.

Question 12 find the second least thickness.

(The visible wavelengths range from 0.4μm to 0.7μm)



Question	n_1	n_2	n_3	Type	L	λ
11.	1.55	1.60	1.33	min	385	

A) 608 nm
B) 567 nm
C) 486 nm
D) 405 nm

FROM THE FORMULA SHEET

$$\lambda = \frac{\lambda_{vacuum}}{n}$$

REMEMBER — not on the formula sheet

Phase table: off a higher n gives $\frac{1}{2}\lambda$, off a lower n gives 0. With no net shift, a transmission minimum needs $2L = \left(m + \frac{1}{2}\right)\frac{\lambda}{n_2}$, $m = 0, 1, 2, \ldots$

Step 1. Phase shifts for r_4: at the 2|3 interface ($1.60 \to 1.33$, off a lower index) the shift is 0, and at the 2|1 interface ($1.60 \to 1.55$, off a lower index) the shift is 0. So the net reflection shift is 0.

Step 2. Minimum condition, solved for λ:

$$\lambda = \frac{2Ln_2}{m + \frac{1}{2}}$$

Step 3. With the key's value $L = 285$ nm: $2Ln_2 = 2(285)(1.60) = 912$ nm.

$$m = 0: \ \frac{912}{0.5} = 1824 \text{ nm}; \qquad m = 1: \ \frac{912}{1.5} = 608 \text{ nm (visible } \checkmark\text{)}; \qquad m = 2: \ \frac{912}{2.5} = 365 \text{ nm (too short)}$$

NOTE ON THE ANSWER KEY

The table prints $L = 385$ nm. With 385 nm, $2Ln_2 = 1232$ nm and the visible minimum is $\dfrac{1232}{2.5} = 493$ nm, which is not an option. The instructor's handwritten solution gets 1824 nm ($m = 0$) and 608 nm ($m = 1$), and those values only come from $L = 285$ nm. So the "385" in the table is a typo. The method above is what matters.

Final answer: **(A) 608 nm** ✓ **Key = A (table typo noted)**

Question 12 Thin film, second-least thickness for a maximum

Question	n_1	n_2	n_3	Type	L	λ
12.	1.68	1.59	1.50	Max	2nd	342

A) 376 nm
B) 269 nm
C) 161 nm
D) 54 nm

FROM THE FORMULA SHEET

$$\lambda = \frac{\lambda_{vacuum}}{n}$$

REMEMBER — not on the formula sheet

With a net $\frac{1}{2}\lambda$ reflection shift, a transmission **maximum** needs $2L = \left(m + \frac{1}{2}\right)\dfrac{\lambda}{n_2}$, $m = 0, 1, 2, \ldots$

Step 1. Phase shifts for r_4: at the 2|3 interface ($1.59 \to 1.50$, off a lower index) the shift is 0. At the 2|1 interface ($1.59 \to 1.68$, off a **higher** index) the shift is $\frac{1}{2}\lambda$. So the net shift is $\frac{1}{2}\lambda$.

Step 2. Maximum condition, solved for L:

$$L = \frac{\left(m + \frac{1}{2}\right)\lambda}{2n_2} = \frac{(2m+1)\lambda}{4n_2}$$

Step 3. The least thickness is $m = 0$: $L = \dfrac{342}{4(1.59)} = 53.8$ nm (option D). The **second** least is $m = 1$:

$$L = \frac{3(342)}{4(1.59)} = \frac{1026}{6.36} = 161 \text{ nm}$$

Final answer: **(C) 161 nm** ✓ **Matches answer key**

Question 13 Non-reflecting acoustic foam

13. An acoustical foam has a density of 20 kg/m^3. What bulk modulus should it have in order to reflect none of the sound normally incident on it? Assume the following values for air: $B_{air} = 1.42\times10^5$ Pa, $\rho_{air} = 1.2$ kg/m^3.

A) $8.5 \times 10^3\ Pa$
B) $5.8 \times 10^3\ Pa$
C) $4.7 \times 10^3\ Pa$
D) $6.5 \times 10^3\ Pa$

FROM THE FORMULA SHEET

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad Z = \sqrt{\rho B}$$

Step 1. No reflection means $r = 0$, so $Z_1 = Z_2$:

$$\sqrt{\rho_{air} B_{air}} = \sqrt{\rho_f B_f} \Rightarrow B_f = \frac{\rho_{air} B_{air}}{\rho_f}$$

Step 2. Numbers:

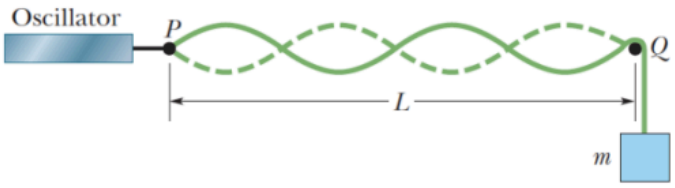
$$B_f = \frac{(1.2)(1.42 \times 10^5)}{20} = \frac{1.704 \times 10^5}{20} = 8.5 \times 10^3 \text{ Pa}$$

Final answer: **(A)** 8.5×10^3 **Pa** ✓ **Matches answer key**

Question 14 Hanging mass for the fifth harmonic

14. In the Figure, a string, tied to a sinusoidal oscillator at P and running over a support at Q, is stretched by a block of mass m. Separation $L = 1.20$ m, linear density $\mu = 1.6$ g/m, and the oscillator frequency $f = 120$ Hz. The amplitude of the motion at P is small enough for that point to be considered a node. A node also exists at Q. What mass m allows the oscillator to set up the **fifth** harmonic on the string?

A) 0.846 Kg
B) 0.752 Kg
C) 0.597 Kg
D) 0.542 Kg



FROM THE FORMULA SHEET

$$f_n = \frac{v}{\lambda_n} = nf_1, \quad v = \sqrt{\frac{F_T}{\mu}}, \quad g = 9.81\ \tfrac{\text{m}}{\text{s}^2}$$

REMEMBER — not on the formula sheet

For a string fixed at both ends, $f_n = \dfrac{nv}{2L}$ (n loops, $\lambda_n = \dfrac{2L}{n}$). The tension from the hanging block is $F_T = mg$.

Step 1. Combine the formulas:

$$f = \frac{n}{2L}\sqrt{\frac{mg}{\mu}} \;\Rightarrow\; m = \frac{4L^2 f^2 \mu}{n^2 g}$$

Step 2. Numbers, with $n = 5$ and $\mu = 1.6 \times 10^{-3}\ \frac{\text{kg}}{\text{m}}$:

$$m = \frac{4(1.20)^2(120)^2(1.6\times10^{-3})}{(5)^2(9.81)} = \frac{132.7}{245.3} = 0.541\ \text{kg} \approx 0.542\ \text{kg}$$

(With $g = 9.8$ you get exactly 0.542 kg.)

Final answer: **(D) 0.542 kg** ✓ Matches answer key

Question 15 Distance where a horn becomes painful

15. At a distance of 10 km, a 100 Hz horn, assumed to be an isotropic point source, is barely audible. At what distance would it begin to cause pain? Table below shows some sound levels.

A) $1.0 \times 10^{-3} m$
B) $1.0 \times 10^{-2} m$
C) $1.0 \times 10^{-4} m$
D) $1.0 \times 10^{-6} m$

Table 17-2 Some Sound Levels (dB)

Hearing threshold	0
Rustle of leaves	10
Conversation	60
Rock concert	110
Pain threshold	120
Jet engine	130

FROM THE FORMULA SHEET

$$I = \frac{P}{4\pi R^2}, \quad \beta = (10\ \text{dB})\log_{10}\left(\frac{I}{I_0}\right), \quad I_0 = 1\times10^{-12}\ \frac{\text{W}}{\text{m}^2}$$

Step 1. Barely audible is 0 dB, so $I_2 = I_0 = 10^{-12}\ \frac{\text{W}}{\text{m}^2}$ at $r_2 = 10^4$ m. Pain is 120 dB, so $I_1 = 10^{-12} \times 10^{12} = 1\ \frac{\text{W}}{\text{m}^2}$.

Step 2. Same power P, so $I \propto \dfrac{1}{r^2}$:

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \;\Rightarrow\; r_1 = r_2\sqrt{\frac{I_2}{I_1}} = 10^4\sqrt{\frac{10^{-12}}{1}} = 10^4 \times 10^{-6} = 1.0\times10^{-2}\ \text{m}$$

Final answer: **(B)** 1.0×10^{-2} **m** ✓ Matches answer key

Question 16 Whistle moving in a circle

16. A whistle of frequency 550 Hz moves in a circle of radius 60 cm at an angular speed of 15.0 rad/s. What is the highest frequency heard by a listener a long distance away, at rest with respect to the center of the circle?

A) 555 Hz
B) 565 Hz
C) 525 Hz
D) 536 Hz

FROM THE FORMULA SHEET

$$f_L = \frac{v \pm v_L}{v \mp v_S} f_S, \quad v_{sound} = 343 \ \tfrac{\text{m}}{\text{s}}$$

REMEMBER — not on the formula sheet

Speed of a point moving in a circle: $v_S = r\omega$

Step 1. Source speed: $v_S = (0.60)(15.0) = 9.0 \ \tfrac{\text{m}}{\text{s}}$. The listener is at rest, so $v_L = 0$.

Step 2. The highest frequency is heard when the whistle moves straight toward the listener (minus sign in the denominator):

$$f_L = \frac{v}{v - v_S} f_S = \frac{343}{343 - 9.0}(550) = \frac{343}{334}(550) = 565 \text{ Hz}$$

Final answer: **(B) 565 Hz** ✓ Matches answer key